Interactive Global Climate Explorer

Team 8

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1 Introduction

We aim to build an interactive climate exploration and classification system for spatial-temporal climate analysis. The project title remains *Interactive Global Climate Explorer*, and the current implementation uses North America as a representative case study. At this stage, we have completed the data pipeline, region-level summaries, climate-feature construction, and a first set of analytical figures.

2 Problem Definition

The central problem is to convert monthly gridded climate data into regional summaries and climate indicators that are reproducible and visually interpretable. For the current milestone, we focus on temperature and precipitation over North America and organize valid grid cells into six macro-regions. The computational goal is reliable monthly, yearly, and climatological statistics; the visual goal is a map where users can inspect regional trends and see a concise explanation of the assigned climate type.

3 Literature Survey

Existing work relevant to our project falls into four main streams.

(1) Koppen-Geiger climate classification. Peel et al. [10], Kottek et al. [8], and Beck et al. [2, 3] provide reliable threshold-based climate classification foundations using monthly temperature and precipitation statistics. Related climate-shift studies such as Rubel and Kottek [12], Rubel et al. [13], and Engelbrecht and Engelbrecht [6] show how these classifications can support broad regional interpretation. These works are scientifically strong, but they are usually distributed as static maps.

(2) Reanalysis and land datasets. Products such as GLDAS [11] and ERA-Interim [5] provide physically consistent meteorological variables suitable for climate analysis, but they require substantial preprocessing before they can be used in an exploratory system.

(3) Statistical climate regime detection. Fovell and Fovell [7], Netzel and Stepinski [9], and Sanderson et al. [14] show that statistical approaches can reveal meaningful climate structure, although these methods are often less interpretable for general users.

(4) Interactive visualization. Shneiderman [15], Andrienko and Andrienko [1], and Bostock et al. [4] provide the main principles for linked spatial-temporal visual analysis. Together, this literature motivates a system that combines interpretable climate statistics with coordinated visual exploration.

4 Proposed Method

Our approach consists of a computation pipeline followed by an interactive visualization layer.

4.1 Computation Pipeline

We implemented a Python workflow that processes monthly climate files sequentially and writes stable outputs in parquet format. The pipeline performs the following steps:

1. Discover all monthly source files and retain only the target analysis window.
2. Load the required monthly temperature and precipitation variables.

3. Remove invalid observations by finite-value filtering.
4. Preserve the original units while deriving temperature in degrees Celsius.
5. Assign each valid grid cell to exactly one macro-region using fixed latitude-longitude rules.
6. Aggregate valid grid cells into region-level monthly summaries.
7. Aggregate region-month summaries into yearly statistics and monthly climatology.
8. Construct interpretable climate indicators for each region.
9. generate figure outputs for downstream visual analysis.

The implementation already produces five analytical tables and processed 72 monthly files for 2020–2025, yielding 5,791,608 valid grid-month observations.

4.2 Key Innovations

- **Streamed grid-table construction** for stable long-format outputs with modest memory use.
- **Transparent macro-regionalization** based on explicit latitude-longitude rules.
- **Multilevel climate summaries** from grid observations to climatology and derived features.
- **Visualization-ready explanation features** such as temperature range and precipitation seasonality.

4.3 Visualization Design

The visualization component is under construction. The main view will be a map of North America in which each region is colored by climate characterization. Hovering over a region will reveal recent temperature and precipitation trends together with a short explanation based on annual mean temperature, annual total precipitation, temperature range, and seasonal precipitation balance. The final interface follows an overview-plus-detail structure: map overview, linked trend panel, and textual evidence box.

5 Experiments and Evaluation Design (under construction)

5.1 Current Results

Current outputs already reveal clear regional contrasts:

- The Southeast is the wettest macro-region, while the Southwest is the driest.
- The South Central and Southeast regions are the warmest on average.
- The North Central region shows the strongest annual temperature range.
- The Southwest shows the strongest precipitation seasonality.

5.2 Upcoming Experiments

The next evaluation stage will focus on:

1. internal consistency checks for all derived statistics;
2. regional comparison analysis based on the climate-feature table;
3. temporal variability analysis from the yearly summaries;
4. interface evaluation for map exploration, trend reading, and explanation quality.

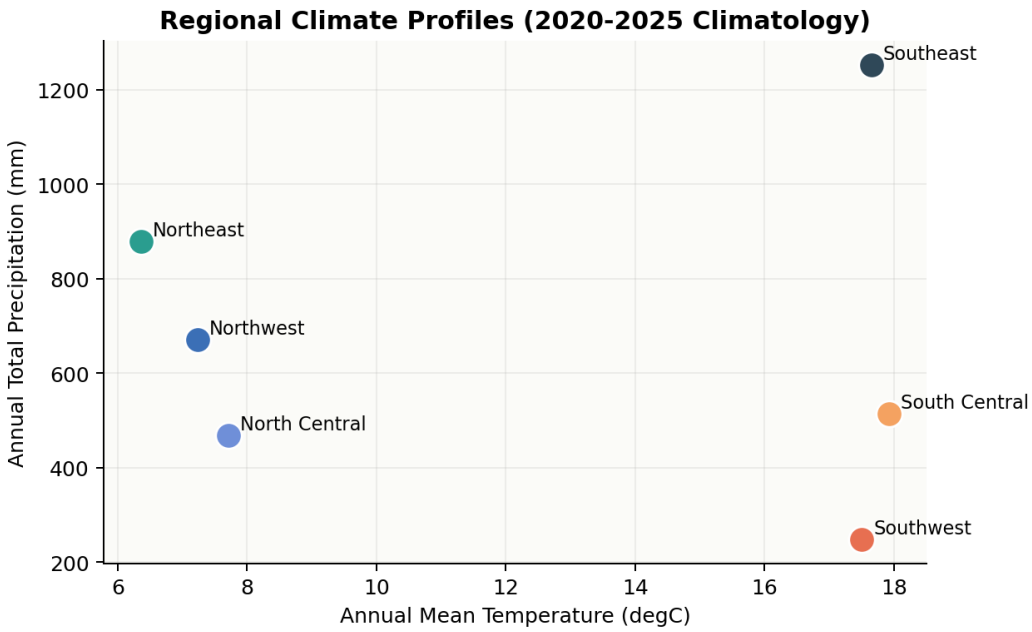

Figure 1: Regional climate profiles derived from the 2020–2025 climatology. The figure summarizes the broad separation among the six macro-regions in annual mean temperature and annual total precipitation.

5.3 Place-holders for Final Results

The final report will add quantitative validation summaries, final interface screenshots, and a concise comparison of regional patterns.

6 Plan of Activities and Team Contributions

6.1 Old Plan

Activity	Member(s)	Time
Topic selection	All	Late February
Data preprocessing	Bohan	Early March
Koppen climate classification	Yi	Mid March
Zonal aggregation	Shuyan	Late March
Interactive visualization	Yuci	Early April
Integration and testing	Jingwei	Mid April
Final report	All	Late April

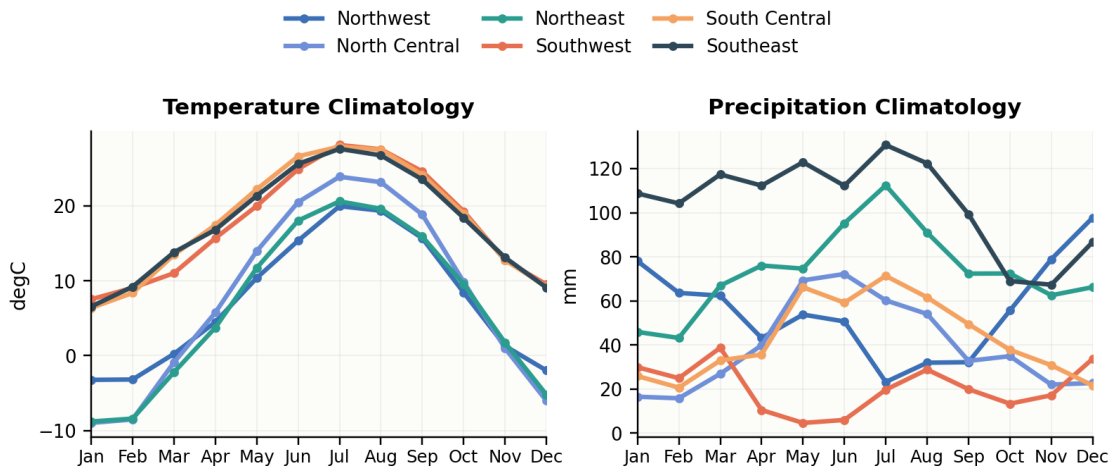

Figure 2: Regional monthly climatology for temperature and precipitation. These trends will be reused in the final interface as the main hover-linked temporal view.

6.2 Current Plan

Phase	Main Work	Status / Time
Data pipeline	Monthly-file reading, valid-cell filtering, schema design, and stable output generation	Completed
Climate summaries	Region-month, region-year, climatology, and climate-feature construction	Completed
Visualization prototyping	Climate map encoding, linked trend views, and explanation-panel specification	In progress
Interactive integration	Hover interaction, coordinated views, and UI assembly	Next phase
Evaluation	Validation checks, regional comparison analysis, and interface testing	Next phase
Final writing	Final screenshots, discussion, and conclusion polishing	Final phase

All team members have contributed a similar amount of effort.

Bohan led data preprocessing and source-file organization. Yi focused on climate-type logic and feature interpretation. Shuyan concentrated on regional aggregation and summary-table design. Yuci focused on visualization concept and interaction design. Jingwei coordinated integration, implementation, testing, and report assembly.

7 Conclusion (under construction)

At this checkpoint, the analytical core of the project is in place. We have completed the climate-processing pipeline, generated stable outputs, derived interpretable regional indicators, and produced prototype visual summaries. The remaining work is focused on completing the interactive interface and final evaluation.

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